

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Global road networks face a dual crisis: **1.6 billion waste tires** are generated annually, with only 40% recycled, while asphalt pavements deteriorate due to aging-induced microcracks that escalate maintenance costs by 30–50% 113. Bitumen aging involves oxidation-triggered **SARA imbalance**, where volatilization of saturates/aromatics increases asphaltene content, causing brittleness and microcracks 13. Traditional solutions (polymer modifiers, synthetic rejuvenators) exhibit limited healing cycles (2–3) and environmental toxicity 29.

Tire waste offers a breakthrough solution:

- **Tire Pyrolytic Oil (TPO)** restores SARA balance via terpenes (e.g., limonene) and monoaromatics, reversing oxidative aging 413.
- **Steel fibers** enable microwave-induced healing, reducing energy needs by 60% versus induction heating 613.
- **Degraded Crumb Rubber (DCR)** enhances elasticity through disrupted cross-linking bonds, enabling autonomous crack closure at 50°C 4.

This research integrates these components into a unified self-healing system—unexplored in existing literature.

1.2 Statement of the Problem

Current self-healing technologies face three critical limitations:

1. **Limited Rejuvenation Cycles:** Capsule-based systems deplete after 1–2 healing events 27.
2. **High Activation Energy:** Induction heating requires >120°C, accelerating bitumen aging 613.
3. **Environmental Toxicity:** Petroleum-based rejuvenators leach polycyclic aromatic hydrocarbons (PAHs) 912.

Tire-derived materials address these gaps but introduce new challenges:

- Untreated TPO contains volatile organic compounds (VOCs)/PAHs 4.
- Conventional Ground Tire Rubber (GTR) reduces mastic stiffness by 30–40%, compromising load-bearing capacity 4.
- No study has optimized *synergies* between TPO, DCR, and steel fibers for multi-mechanism healing.

1.3 Objectives

General Objective:

To develop a tire-integrated self-healing asphalt system combining pyro-rejuvenation, elastic recovery, and microwave activation.

Specific Objectives:

1. Formulate TPO by distilling the 160–200°C fraction to maximize limonene (33.7 wt%) and minimize PAHs 412.
2. Engineer DCR (Mooney viscosity: 0–80) functionalized with silane to enhance bitumen compatibility 4.
3. Design steel fiber-reinforced asphalt (0.3% wt) with 10–15mm lengths for microwave absorption 6.
4. Quantify healing efficiency via neutron imaging (crack volume reduction) and dynamic shear rheometer (DSR) fatigue–healing–fatigue tests 13.

1.4 Significance of the Study

Table 1: Stakeholder Impact Analysis

Stakeholder	Impact
Environmental Agencies	Diverts 1.5 tons of waste tires per 10m ³ asphalt; reduces CO ₂ by 60% vs. virgin aggregates 912
Road Authorities	Lowers maintenance costs by 50% via 5+ healing cycles 613
Developing Economies	Enables low-energy healing (solar/microwave) without industrial equipment 612
Research Community	Advances knowledge of sulfur-bond re-crosslinking in DCR for low-temperature healing 4

1.5 Scope and Limitations

Scope:

- **Materials:** CA-24 bitumen; TPO from waste tires; steel slag aggregates (80% replacement) 6.
- **Healing Triggers:**
 - *Autonomous:* Solar heating (>50°C) for TPO diffusion.
 - *Active:* Microwave (2.45 GHz, 700W, 60s) for steel fibers 6.
- **Testing:**
 - Laboratory-scale specimens (100mm diameter × 63.5mm height).
 - Field validation on low-traffic roads (e.g., campus driveways).

Limitations:

- **Material Variability:** TPO distillation efficiency depends on tire composition 4.
- **Scale Effects:** Industrial compaction (e.g., roller) not replicated in lab manual ramming 6.
- **Climate Constraints:** Field trials limited to tropical climates (25–35°C).

CHAPTER 2: LITERATURE REVIEW

2.1 Intrinsic Healing Mechanisms

Bitumen self-heals via **viscoelastic flow** and **molecular diffusion**, governed by temperature and rest periods. Wool’s reversible crack model (Figure 1) illustrates this process, where capillary action draws bitumen into microcracks, followed by molecular entanglement 13. However, ambient conditions limit crack closure to <50µm without additives 13. Key factors influencing intrinsic healing include:

- **Molecular Weight:** Long-chain saturates diffuse 40% faster than asphaltenes 13.
- **Temperature:** Healing efficiency doubles between 20°C and 50°C but plateaus above 85°C 613.
- **Aging:** Oxidized bitumen requires aromatics replenishment—achievable via TPO’s 52.9% monoaromatics 413.

2.2 Extrinsic Healing Technologies

Table 2: Comparative Analysis of Self-Healing Methods

Method	Healing Agent	Activation	Efficiency	Limitations
Microcapsules	Sunflower oil	Crack rupture	60–70% (1 cycle)	Single-use; low survival during mixing 7
Induction	Steel fibers	Magnetic field	80% (3 cycles)	High energy cost (>120°C); requires metal additives 6
TPO Rejuvenation	Distilled tire oil	Heat >50°C	95% (5 cycles)	VOC emissions if undistilled 4

2.2.1 Microwave-Activated Systems

Steel fibers (0.2–0.4% wt) convert microwaves to localized heat, enabling crack repair at 70°C—50% lower than induction 6. Studies show:

- Steel slag aggregates enhance thermal uniformity, achieving 92% healing versus 78% with granite 6.

- Graphite powder (15% wt) increases microwave absorption but reduces tensile strength by 20% 6.

2.2.2 Rejuvenator-Based Systems

Microvascular networks (e.g., PVDF tubes) release rejuvenator upon cracking:

- Calcium-alginate capsules withstand mixing temperatures up to 180°C 7.
- Optimal dosage: 3 microvasculars/cm² achieves 86% healing efficiency after two cycles 7.

2.3 Tire-Derived Materials in Asphalt

2.3.1 Tire Pyrolytic Oil (TPO)

- **Rejuvenation Mechanism:** Limonene in TPO (160–200°C fraction) swells aged bitumen, reducing viscosity by 300% 413.
- **Environmental Advantage:** Distillation eliminates 90% PAHs, aligning with EPA Method 1311 412.

2.3.2 Degraded Crumb Rubber (DCR)

- **Elastic Recovery:** Pre-desulfurized DCR (Mooney viscosity: 0) forms nano-networks that bridge cracks at low temperatures 4.
- **Self-Healing Synergy:** Dynamic disulfide bonds in DCR enable re-crosslinking, enhancing healing at 10–25°C 4.

2.3.3 Steel Fibers

- **Microwave Susceptibility:** 10–15mm fibers heat to 70°C within 60s at 700W 6.
- **Dispersion Optimization:** Silane functionalization prevents clustering and maintains air void content <4% 6.

2.4 Knowledge Gaps

1. **Hybrid System Synergy:** No studies combine TPO diffusion, DCR elasticity, and microwave activation 413.
2. **Low-Temperature Performance:** Healing below 20°C remains unexplored for tire-asphalt composites 4.
3. **Toxicity of Degraded Components:** Leachate risks from heavy metals (Zn, Cd) in steel fibers require EPA 1311 validation 613.

CHAPTER 3: METHODOLOGY

3.1 Material Design and Preparation

Table 3: Tire-Derived Components and Processing

Component	Source	Treatment	Function
TPO	Waste tire pyrolysis	Distillation (160–200°C cut)	Replenishes lost saturates/aromatics
DCR	Cryo-ground tire crumb	Silane functionalization	Enhances elasticity; microcrack bridging
Steel Fibers	Tire wire recovery	Cutting (15mm); acid cleaning	Microwave absorption; crack bridging

Asphalt Formulation:

- Base Mix: AC-13 gradation with 4.9% virgin bitumen 6.
- Modifications:
 - 6% TPO (rejuvenator)
 - 25% DCR (filler replacement)
 - 0.3% steel fibers
 - 80% steel slag aggregates (replacing virgin aggregate) 6.

3.2 Experimental Workflow

Diagram

Code

3.3 Key Experiments

1. Healing Efficiency Quantification:

- **Neutron Tomography:** 3D crack volume measurement pre/post-healing (detection limit: 5µm) 13.
- **Healing Index (HI):** $HI = (\text{Post-heal strength} / \text{Original strength}) \times 100$ 6.
- **Fatigue-Healing-Fatigue Test:** Four-point bending to 50% modulus reduction, followed by healing and retesting 13.

2. Mechanical Performance:

- **Rutting Resistance:** Asphalt Pavement Analyzer (APA) at 60°C (10,000 cycles @ 0.7 MPa) 6.
- **Moisture Sensitivity:** Tensile Strength Ratio (TSR) after AASHTO T283 freeze-thaw 13.

- **Wet Grip:** British Pendulum Tester (PTV > 70 target) 6.

3. **Environmental Impact:**

- **Leachate Analysis:** EPA Method 1311 for heavy metals (Zn, Cd) 4.
- **Life Cycle Assessment (LCA):** SimaPro v9.0 comparing cradle-to-gate emissions vs. conventional asphalt 12.

3.4 Data Analysis

- **Machine Learning:** Gaussian process regression to optimize TPO/DCR/fiber ratios for max HI.
- **Statistical Validation:** ANOVA (p<0.05) for healing parameters (temperature, rest period).

3.5 Feasibility for Student Implementation

Table 4: Low-Cost Equipment Alternatives (Philippine Context)

Equipment	Standard Version	Student Adaptation	Cost (₱)
Pyrolysis Reactor	Industrial rotary kiln	Modified charcoal kiln + condenser	15,000
Compactor	Gyratory compactor	Manual rammer (50 blows/side)	500
Heating Source	Microwave chamber	Household microwave (700W)	3,000
Binder Testing	DSR	Ductility test at 10°C	Lab-owned

Safety Protocols:

- TPO distillation: Ventilated fume hood + activated carbon filters.
- Microwave activation: RF-shielded enclosures.

Innovative Contributions

1. **Triple-Action Healing:**

- **Molecular Rejuvenation** (TPO → SARA rebalance).
- **Elastic Recovery** (DCR → sulfur-bond re-crosslinking).
- **Thermal Activation** (Steel fibers → microwave absorption).

2. **Waste Valorization:** 95% tire utilization (fibers, rubber, oil).

3. **Low-Energy Healing:** Solar/microwave activation feasible in off-grid regions.

Expected Outcomes

- Healing Index $\geq 80\%$ after 5 cycles.
- Wet grip PTV > 75 (25% improvement vs. conventional asphalt).
- 50% cost reduction vs. polymer-modified self-healing systems.

Next Steps: Chapter 4 will detail field validation protocols, including IoT-sensor deployment for real-time crack monitoring. Appendices will include material safety datasheets for distilled TPO.